

Day 27 — Confusion Matrix and Threshold Tuning

Veda Technology · Data Science Track · Level 2 · Day 27

Objective: Understand how probability thresholds affect classification performance.

1. Dataset

Breast Cancer Wisconsin Dataset from scikit-learn. Target labels: 0 = malignant and 1 = benign.

2. Workflow

- Load data
- Train/test split and scaling
- Train Logistic Regression
- Use `predict_proba()`
- Create confusion matrix
- Test multiple thresholds
- Compare precision, recall and F1
- Recommend threshold using highest tested F1-score

3. Core Code

```
probabilities = model.predict_proba(X_test_scaled)[: , 1]

for t in [0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80]:
    pred = (probabilities >= t).astype(int)
    precision = precision_score(y_test, pred)
    recall = recall_score(y_test, pred)
    f1 = f1_score(y_test, pred)

best = threshold_results.loc[threshold_results["f1"].idxmax()]
```

4. Interpretation

Lower thresholds generally increase recall; higher thresholds generally increase precision. The appropriate threshold depends on the relative cost of false negatives and false positives.

5. Interview Questions

What is a confusion matrix? A table of TP, TN, FP and FN. **How does threshold affect recall?** Lowering it generally increases positive predictions and recall. **Why not always 0.5?** Because the best threshold depends on the application and error costs.

6. Conclusion

Threshold tuning allows a classification model to be adjusted for the desired precision-recall trade-off instead of blindly using 0.5.